

Project Report

Restaurant Management System using C#

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Course Name & Code : Object Oriented Programming
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1. Objective

The main objective of this project is to design and implement a console-based Restaurant Management System using C# that demonstrates the practical use of Object-Oriented Programming (OOP) concepts.

This system allows:

- Customers to view menu items
- Add items to a cart with quantity selection
- Remove items from the cart
- Calculate total bill including VAT
- User login, signup, and logout
- Admin panel to manage menu and view customer orders

The project is developed to strengthen understanding of classes, inheritance, abstraction, constructors, operator overloading, static members, and arrays in C#.

2. Review and Limitations

Review-

The system successfully simulates a small restaurant ordering process using a menu–cart–checkout flow. It supports multiple users, maintains individual carts, and provides an admin panel for menu and order monitoring.

The design follows a modular approach, making the code readable and extendable.

Limitations-

- Data is not stored permanently (no database or file handling).
 - Usernames and passwords are stored in arrays, not encrypted.
 - The application is console-based, no graphical user interface (GUI).
 - No payment gateway integration.
 - Limited number of users and menu items due to fixed array sizes.
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3. Methods Used and Their Features

Key Classes and Methods-

Greet Class

(Coded By 2243081066)

- greeting()
 - Displays welcome message.

MenuP (Abstract Class)

(Coded By 2243081066)

- print()
 - Forces subclasses to implement menu display logic.

MenuItem Class

(Coded By 2243081066)

- Constructors (normal, overloaded, copy constructor)
- print()
 - Displays menu item details.
- Demonstrates inheritance and abstraction.

OrderItem Class

(Coded By 2243081066)

- Stores selected menu item and quantity.
- Operator overloading (+)
 - Merges quantities of the same item.

Cart Class

(Coded By 2243081074)

- add(MenuItem obj, int n)
 - Adds item with quantity.
- add(MenuItem obj)
 - Overloaded method.
- showcarr()
 - Displays current cart.
- removeitem(int r)
 - Removes selected item.
- Total()
 - Calculates total price.
- static method Vatcal(double v)
 - Calculates VAT.

- empty()
 - Clears cart after checkout.

Usersystem Class

(Coded By 2231081075)

- signup()
 - Registers a new user.
- login()
 - Logs in existing user.
- logout()
 - Logs out without deleting user data.
- Maintains user-specific carts.

AdminP Class

(Coded By 2243081121)

- panel()
 - Admin login
 - View customers
 - View all orders
 - Add/remove menu items

Program Class

(Coded By 2243081075)

- main()
 - It initializes essential objects such as:
 - MenuItem[] for storing restaurant menu
 - Cart for managing selected items
 - Usersystem for handling login, signup, and logout
 - Calls appropriate methods based on user selection, such as:
 - Showing menu items
 - Adding or removing cart items
 - User login, signup, and logout
 - Checkout and bill calculation
 - Accessing the admin panel
 - The Main() method effectively integrates all components of the system and ensures smooth execution of restaurant operations.
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4. Output

Sample Output Flow-

- This is the initial home page of the Project, where user can select multiple options available.

```
-----WELCOME TO C# RESTAURANT!-----  
      Please select the options below  
  
=====OPTIONS=====  
1. Show Menu  
2. View Cart  
3. Remove Item  
4. Check Out  
5. Exit  
6. Login / Signup  
Select Option : █
```

- To order customers desired item they must select option 1 to view the menu and select their desired item with their desired quantity.

```
Select Option : 1  
  
=====MENU=====  
1.    Burger           -150TK  
2.    Pizza            -350TK  
3.    Pasta            -180TK  
4.    Rice Bowl        -170TK  
5.    Cold Coffee      -120TK  
6.    Soft Drinks      -30TK  
=====
```

Item ID	Item Name	Price (TK)
1	Burger	150
2	Pizza	350
3	Pasta	180
4	Rice Bowl	170
5	Cold Coffee	120
6	Soft Drinks	30

```
Enter Item ID To Add(0 to go back) : █
```

- After selection the selected items will be automatically added to cart, by selecting the option 2 customer can see what items they selected.

```
Select Option : 2

===== (YOUR CART) =====
1.Burger      x2  = 300TK
2.Pizza       x1  = 350TK
=====
```

- When customer want to remove any selected item from their cart, they can do so by selecting the option 3 and then choosing the item that they want to discard.

```
Select Option : 3

===== (YOUR CART) =====
1.Burger      x2  = 300TK
2.Pizza       x1  = 350TK
=====
Enter item number to Remove : 2
Pizza         is removed
```

- After getting satisfied with the order items customer must create an account or sign in with an existing account before checking out.

```
Select Option : 6
-----
1| Log-in
2| Sign-up
3| Admin-Panel
4| Log-out
Select an option to proceed(0 to go back) : 2
```

- When customer is logged in by selecting option 4 they can check out and see their total bill.

```
Select Option : 4

===== (YOUR CART) =====
1.Burger      x2  = 300TK
=====
-----* FINAL BILL *-----
      Total      : 300
      VAT-(5%)   : 15
      -----
      Sub-Total  : 315
      -----
Thank you User1 for visiting us!!!
Cart has been refreshed.
```

- There is a separate admin control panel which is in option 6 then selecting third option. In here admin can view customer details and their orders, add or remove menu items.

```
Enter admin username : admin
Enter admin password : 1075
Admin Login Successful!

===== ADMIN PANEL =====
1. View All Customers
2. View Cart (Orders)
3. Add Menu Item
4. Remove Menu Item
0. Go Back
Select Option : █
```

5. Conclusion

This project successfully demonstrates how a Restaurant Management System can be implemented using core C# and OOP principles.

The use of abstraction, inheritance, method overloading, operator overloading, and static members makes the program structured and reusable.

Although the system has limitations such as lack of database support and GUI, it effectively fulfills the learning objectives of an academic project. With further enhancement, this system can be extended into a fully functional restaurant application with database connectivity and graphical interface. The project meets its objectives and provides a solid foundation for understanding real-world application development using C#.
